

# VU MECHANICS – LAWI 100515 (PI)

## SS 2026

### 2nd Exercise Exam

26.06.2026 GROUP A

|                     |          |
|---------------------|----------|
| Name:               |          |
| Student ID number:  |          |
| Study program code: |          |
| Notes               |          |
| Result              | ____/100 |

| Tasks                                  | Possible Points | Achieved Points |
|----------------------------------------|-----------------|-----------------|
| 1) Task 1 – Section Forces and Moments | 40              |                 |
| 2) Task 1 – Truss System               | 30              |                 |
| 3) Task 2                              | 30              |                 |

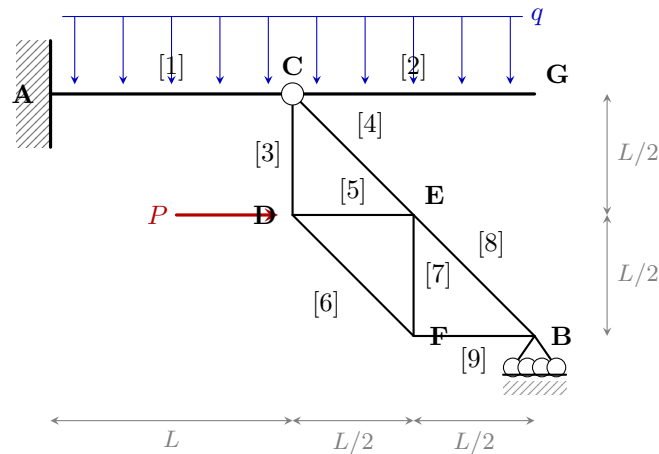
**Working Time:** 90 min **Passing Grade:**  $\geq 50\%$ **Permitted Aids:** Formula sheet (boku-learn), calculator, set square, compass. No pencil, no red pen, no own paper.

## 1. Task

(70 %)

**Given** is the planar static system shown below. Dimensions and system parameters are given in the sketch. The structure is loaded by a **uniformly distributed load**  $q$  acting on beams [1] and [2], and a **concentrated load**  $P$  at truss node D.

|                     |                      |                     |
|---------------------|----------------------|---------------------|
| $L = 1.5 \text{ m}$ | $q = 3 \text{ kN/m}$ | $P = 15 \text{ kN}$ |
|---------------------|----------------------|---------------------|



## Tasks:

1. Check whether the system is statically determinate.
2. Determine the support reactions **in general form as functions of  $q$ ,  $P$ , and  $L$** . Draw the reactions into the system sketch.
3. Evaluate the support reactions for the given parameters:

$A_V =$

$A_H =$

$M_A =$

$B_V =$

4. Determine the section forces  $M(x)$ ,  $V(x)$ , and  $N(x)$  in beams [1] and [2] **in general form as functions of  $q$ ,  $P$ , and  $L$** . Define the range of the running variable  $x$ . Evaluate the functions at the beam start and end:

| Beam start |          |          | Beam end  |          |          |
|------------|----------|----------|-----------|----------|----------|
| $M$ [kNm]  | $V$ [kN] | $N$ [kN] | $M$ [kNm] | $V$ [kN] | $N$ [kN] |
| Beam [1]   |          |          |           |          |          |
| Beam [2]   |          |          |           |          |          |

5. Draw the distributions  $M(x)$ ,  $V(x)$ , and  $N(x)$  and mark the characteristic values.
6. Calculate the axial forces  $S_8$  and  $S_9$  using a **circular cut (Rundschnitt)** at node B, in general form as a function of  $P$ :

$S_8 =$

$S_9 =$

7. Calculate the axial forces  $S_4$ ,  $S_5$ , and  $S_6$  using **Ritter's cut (Ritterschnitt)**, in general form as a function of  $P$ :

$S_4 =$

$S_5 =$

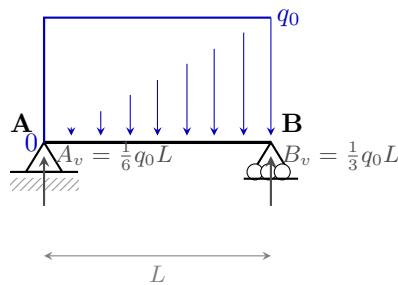
$S_6 =$

**2. Task****(30 %)**

**Given** is a simply supported beam (span  $L$ ) with a triangular load (0 at A,  $q_0$  at B). The support reactions are  $A_v = \frac{1}{6}q_0L$  and  $B_v = \frac{1}{3}q_0L$ . The bending moment distribution is:

$$M(x) = -\frac{q_0}{6L}x^3 + \frac{q_0L}{6}x \quad (0 \leq x \leq L)$$

|                     |                        |
|---------------------|------------------------|
| $L = 1.5 \text{ m}$ | $q_0 = 8 \text{ kN/m}$ |
|---------------------|------------------------|

**Tasks:**

1. Determine the shear force  $V(x)$  as a function of  $x$ .
2. Find the position  $x_{\max}$  along the beam where the moment is maximum.
3. Calculate the corresponding maximum moment  $M_{\max}$ .
4. Sketch the distributions of  $V(x)$  and  $M(x)$  with all characteristic values.